

Telco Customer Churn: Predictive Retention Analytics

End-to-End Data Analytics Project | SQL • Python • Power BI/Tableau • Statistical Analysis

1. Executive Summary

The telecommunications industry operates in a saturated, highly price-sensitive market where the cost of acquiring a new customer is estimated to be 5-7x higher than retaining an existing one. This project simulates a real-world analytics engagement for a fictional telecom provider, **Telco Inc.**, which is experiencing an elevated customer churn rate (~26.5%) that is eroding recurring revenue. Leadership currently has no systematic way to identify which customers are at risk of leaving or why — decisions are made reactively, after revenue has already been lost.

This project delivers an end-to-end analytics workflow — from raw data to a stakeholder-ready dashboard — that identifies churn drivers, segments at-risk customers, and translates findings into concrete retention recommendations for the business.

2. The Core Problem

The company lacks a proactive, data-driven strategy to identify at-risk customers before they cancel service. Retention efforts today are reactive: churn is only confirmed after a customer has already left, at which point the revenue and acquisition cost are already sunk. The business needs to shift from a **reactive** posture (discovering churn after the fact) to a **predictive, insight-driven** posture (flagging at-risk segments early enough for the retention team to intervene).

3. Business Objectives

The primary goal is to reduce customer churn by equipping the retention and marketing teams with clear, actionable, data-backed insights. Specific objectives:

- **Identify Key Drivers:** Quantify which factors (contract type, monthly charges, tenure, payment method, lack of tech support/security add-ons, internet service type) correlate most strongly with churn.
- **Profile the At-Risk Customer:** Build clear demographic and behavioral segments of customers most likely to cancel, using cohort and RFM-style analysis.
- **Quantify Revenue at Risk:** Estimate Customer Lifetime Value (CLV) proxies and the monthly recurring revenue exposed to churn within each risk segment.
- **Recommend Strategic Actions:** Provide specific, data-backed recommendations on pricing, contract restructuring, and service bundling to improve retention.
- **Enable Ongoing Monitoring:** Deliver an interactive dashboard so stakeholders can self-serve churn insights without rerunning the analysis manually.

4. Dataset

This project uses the publicly available **IBM/Kaggle Telco Customer Churn dataset** (~7,043 customers, 21 attributes), which closely mirrors real telecom operational data. Attributes span four categories:

Category	Example Fields
Demographics	Gender, senior citizen status, partner, dependents
Account Info	Tenure, contract type, paperless billing, payment method, monthly & total charges
Services Subscribed	Phone, multiple lines, internet service, online security/backup, tech support, streaming TV/movies
Target Variable	Churn (Yes/No)

5. Analytics Scope & Methodology

5.1 Data Cleaning & Exploratory Data Analysis (Python)

- Handle missing/incorrect data types (e.g., blank TotalCharges for new customers), duplicates, and outliers.
- Engineer features such as tenure buckets, average monthly spend, and services-subscribed count.
- Perform univariate and bivariate analysis to surface initial churn patterns using pandas, matplotlib, and seaborn.

5.2 In-Depth SQL Analysis

- Write queries to calculate churn rate by contract type, tenure cohort, payment method, and service bundle.
- Build CLV proxy metrics (tenure × monthly charges) to rank customer segments by revenue value.
- Segment and rank high-risk, high-value customers who should be prioritized for retention outreach.

5.3 Statistical Validation

- Use chi-square tests for categorical variables and correlation analysis for numeric variables to confirm which drivers are statistically significant rather than coincidental.

5.4 Dashboard & Reporting

- Build an interactive Power BI / Tableau dashboard summarizing churn KPIs, at-risk segments, and revenue impact.
- Prepare a one-page executive summary and slide deck translating findings into three prioritized business recommendations.

6. Tools & Tech Stack

Layer	Tools
Data Cleaning & EDA	Python (pandas, NumPy, matplotlib, seaborn) / Jupyter Notebook
Data Querying	SQL (PostgreSQL / MySQL)
Visualization & Dashboarding	Power BI or Tableau

Version Control	Git & GitHub (for reproducibility and portfolio presentation)
-----------------	---

7. Key Deliverables

- Cleaned dataset + documented Python EDA notebook.
- SQL query bank answering the business's key churn questions (churn rate by segment, CLV, at-risk revenue).
- Interactive Power BI/Tableau dashboard for ongoing self-serve monitoring.
- One-page executive summary with three prioritized, data-backed retention recommendations.
- Public GitHub repository with README documenting problem, approach, and findings.

8. Success Metrics (KPIs)

- Overall churn rate and churn rate variance across key segments (contract type, tenure, service bundle).
- % of total revenue concentrated in identified high-risk segments.
- Number of statistically validated churn drivers identified (via hypothesis testing).
- Clarity/actionability of final recommendations, as judged against stated business objectives.

Prepared as a self-initiated, portfolio-grade data analytics case study | Dataset: IBM/Kaggle Telco Customer Churn